Steering Microbial Communities with an Energy-Based JEPA World Model

Inverse Dynamics, Metric Geometry, and the Limits of Planning in Latent Space

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Abstract

We study energy-based Joint-Embedding Predictive Architectures (JEPA) as *world models of living systems under intervention*: given the state of a microbial community and an applied perturbation (a dose on a candidate-species panel), predict the next state *entirely in latent space*, never reconstructing abundances. Building on the `eb_jepa` framework, we contribute (i) a permutation-invariant set-transformer encoder that consumes a community as an unordered set of species tokens; (ii) a generalized Lotka–Volterra (gLV) "Two-Rooms" simulator whose cyclic inter-guild competition makes target states reachable only through *non-monotonic* intervention sequences; and (iii) an action-conditioned latent world model with an inverse-dynamics (IDM) auxiliary. Across controlled, seed-replicated experiments we report a mix of positive, negative, and diagnostic results, all measured. (1) In a collapse-prone regularization regime, the IDM term rescues the decodability of the applied intervention from the latent state (R^2: $0.520 \to 0.748$, $+0.229$, positive in all three seeds), a textbook slow-feature collapse-and-recovery; the effect is regime-dependent and shrinks under strong variance regularization. (2) On a MDSINE2-style hold-one-subject-out temporal benchmark the JEPA world model is the strongest forecaster at every horizon. (3) On a real infant-gut downstream task a frozen JEPA embedding with a SIGReg (LeJEPA) regularizer matches or beats a published baseline. (4) Planning is a *diagnosed negative and its closure*: a pure-JEPA latent is faithful for one-step prediction yet is *not a metric space*, so latent model-predictive control succeeds 0% of the time across eight independent levers; supplying a single isometry auxiliary turns planning $0\% \to 100\%$ at oracle quality (final distance 0.804 vs. oracle 0.790), and this closure generalizes to 4/4 held-out gLV instances. (5) Removing sequencing-technology nuisance is hard: an adversary (DANN) fails while a deterministic moment-matching alignment (CORAL) gives a partial, linear-only fix. We close with a real-world generality check on Tahoe-100M drug perturbations. Our central lesson: *faithful one-step prediction does not imply a plannable metric*, and the metric must be supplied, not merely hoped for.

Reproducibility & integrity. Every quantity below is labelled [M] MEASURED (from a committed run, with job/figure provenance) or [E] EXPECTED (a hypothesis or literature value). Negative and surprising results are reported as-is. The single load-bearing integrity correction (an unseeded-decoder fluke) is documented in Section 6. Code, configs, result JSONs, and the figures reproduced here live in `examples/microbiome_jepa/`, `examples/microbiome_benchmark/`, and `examples/tahoe_probe/`; a complete quantitative log is in `docs/WORK_REPORT.md`.

1 Introduction

Microbial communities — the gut, soil, water — are high-dimensional dynamical systems we increasingly wish to *control*: nudge a dysbiotic gut toward a healthy configuration, design a probiotic intervention, or screen a drug's effect on a cell. Control requires a *world model*: a map from (state, action) to the next state. We ask whether the Joint-Embedding Predictive Architecture (JEPA) family [1] — which predicts in a learned representation space rather than input space, under an energy/anti-collapse objective — yields world models that are good enough not just to *predict* a community's trajectory but to *plan* interventions on it.

JEPA is attractive here for two reasons. First, microbial state is compositional, sparse, and permutation-invariant (a community is a *set* of species, not an ordered vector); predicting in latent space sidesteps the pathologies of reconstructing zero-inflated counts. Second, a learned latent provides a natural substrate for energy-based planning: descend the predicted energy landscape toward a target embedding. But JEPA also has a well-known failure mode — it preferentially encodes *slow* features and can collapse away the fast, action-driven signal we most need for control [3].

This paper is an honest, exhaustive study of where this works and where it does not. We build the necessary machinery (a set encoder, a controllable simulator, an action-conditioned predictor with an inverse-dynamics term), establish positive results on representation quality and short-horizon forecasting, and then confront planning head-on. The planning story is the heart of the paper: a pure-JEPA latent that is demonstrably faithful for one-step prediction nonetheless fails completely as a *planning metric*, and we trace the bottleneck layer by layer to the geometry of the representation — then close the loop with a single, well-understood auxiliary.

Contributions.

- A permutation-invariant **set-transformer encoder** for microbial communities that plugs into `eb_jepa`'s temporal predictor and regularizers unchanged (Section 3.2).
- A **gLV "Two-Rooms" simulator** with cyclic, rock-paper-scissors inter-guild competition, giving provably non-monotonic reachability — a clean, controllable testbed for planning (Section 3.4).
- A measured **IDM collapse-and-recovery** result: the inverse-dynamics auxiliary restores fast (action) decodability in the collapse-prone regime, and the effect is regime-dependent (Section 4.1).
- A **layered diagnosis of planning** (controllability → representation geometry → readout fidelity) showing that *faithful prediction* $\neq$ *plannable metric*, and a **metric (isometry) auxiliary** that closes the planning loop 0% → 100% and generalizes across instances (Section 4.4).
- Studies of **anti-collapse regularizers** (VICReg vs. SIGReg/LeJEPA) and of **sequencing-technology invariance** (adversarial vs. moment-matching), plus a real-world generality check on **Tahoe-100M** drug perturbations (Sections 4.3–4.6).

2 Background and Related Work

JEPA and energy-based SSL. JEPA [1] predicts the representation of a target from a context, scoring candidates by an energy $\mathcal{E}(x, x') = \|g_\phi(f_\theta(x)) - f_\theta(x')\|^2$. To avoid the trivial constant solution, an anti-collapse regularizer is added; we use VICReg's variance–covariance terms [2] and, as an alternative, the SIGReg isotropy objective of LeJEPA [4], an Epps–Pulley characteristic-function test that pushes the embedding toward an isotropic Gaussian. A world model is the action-conditioned case: the predictor $g_\phi(f_\theta(x), q_\omega(a))$ takes an encoded action a.

Slow-feature collapse. Sobal et al. [3] show that a temporal JEPA trained only to predict the next latent tends to encode *slow* features (trajectory identity/basin) and discard the *fast* signal (the applied action). The standard remedy is an inverse-dynamics model (IDM): require that the action be recoverable from consecutive latents, forcing the encoder to keep fast information. This is the mechanism we test directly in Section 4.1.

Microbiome dynamics and planning. Generalized Lotka–Volterra (gLV) models are the workhorse of microbial ecology; MDSINE2 [5] formalizes inference and hold-one-subject-out forecasting on real time series. For control we use model-predictive path integral (MPPI) control [6], sampling action sequences and reweighting by a cost. Our nuisance-removal study draws on domain-adversarial training (DANN) [7] and deep moment-matching (CORAL) [8]. The closest learned-representation neighbour for cellular perturbations is GeneJEPA [9]; we compare our own baselines cleanly rather than claiming to beat its protocol.

3 Methods

3.1 Problem setup

A community state at time t is a set $\{s_i\}$ of present species, each with an embedding and a log-abundance; the action $a_t \in \mathbb{R}^K$ is a non-negative dose applied to a fixed panel of K candidate species. The world model is an energy

$$\mathcal{E} \;=\; \big\| \, g_\phi\big(f_\theta(x_t),\, q_\omega(a_t)\big) - f_\theta(x_{t+1}) \, \big\|^2 \;+\; \lambda \, R(z), \tag{1}$$

with f_θ the encoder, g_ϕ the action-conditioned predictor, q_ω the action encoder, and R an anti-collapse regularizer. At inference we are given (x_t, a_t) and read off $\hat{z}_{t+1}$; planning optimizes a sequence $a_{t:t+H}$ so that the rolled-out latent reaches a target embedding.

3.2 Permutation-invariant set-transformer encoder

The encoder (`SetTransformerEncoder`, in `eb_jepa/architectures.py`) maps an observation dict `{otu:[B,T,N_max,F], mask:[B,T,N_max]}` to a latent of shape exactly $[B, D, T, 1, 1]$, matching the convolutional-encoder convention so that the temporal predictor and regularizers work unchanged. Each token is `concat(species_embedding (384-d), CLR log-abundance)`, F=385. Tokens are embedded by a single linear map with *no positional encoding*, passed through a `TransformerEncoder` with a padding mask, and pooled by a masked mean (or a learned single-query attention pool). Both pooling choices are order-independent, so the encoder is permutation-invariant by construction (verified to $< 10^{-5}$ max-abs deviation in a smoke test).

3.3 Data pipeline

A centered log-ratio (CLR) transform removes the sum-to-one compositional constraint; a per-dimension z-score (fit on the training corpus, with a floor on near-constant dims) prevents the abundance channel from dwarfing the 384 embedding dims in the variance term. Two-view SSL augmentations (species subsampling, dropout, log-abundance jitter) and I-JEPA-style masking are provided. Normalization is not cosmetic: per-feature z-scoring breaks a temporal collapse and is the single largest lever on downstream forecasting skill in our ablations.

3.4 The gLV "Two-Rooms" simulator

The simulator (`eb_jepa/datasets/microbiome/glv.py`) integrates $\dot{x} = x \odot (r + Ax) + m$ by RK4 with non-negativity clamping. Species are partitioned into n_{guilds} competitive guilds with weak within-guild mutualism and *cyclic* between-guild competition (guild g strongly suppresses g+1). Competitive exclusion makes each single-guild-dominant corner a stable attractor (verified by a negative-real-part Jacobian). The cycle makes reachability *non-monotonic*: steering "against the cycle" requires first blooming a gate guild — moving the distance-to-target *up* before down. A greedy "reduce distance every step" policy fails on all 6/6 ordered attractor pairs [M]; a detour succeeds. This is the microbiome analogue of the classic "Two Rooms" planning task and the reason the testbed is a meaningful control problem rather than a smooth descent.

3.5 Action-conditioned world model and inverse dynamics

The Layer-B world model wraps gLV trajectories into `eb_jepa`'s `TrajDataset` 5-tuples and trains the recurrent predictor with the `VC_IDM_Sim_Regularizer`: VICReg variance/covariance terms, a temporal similarity term, and the inverse-dynamics term `idm_coeff`, which trains an MLP IDM$(z_t, z_{t+1}) \to \hat{a}_t$ and back-propagates through the encoder. `idm_coeff` is our principal ablation knob. We also implement a `SIGReg_IDM_Sim_Regularizer` that swaps VICReg's variance/covariance for SIGReg isotropy on the encoder output.

3.6 Planning: latent MPPI and the metric auxiliary

Planning uses MPPI: sample N action sequences over horizon H, roll out the latent predictor, score each by a cumulative cost (latent ℓ_2 distance to the target embedding, or a decoded-state distance, or a learned monotonic cost head), reweight, and iterate. The *metric auxiliary* (the HYBRID variant) adds a term that asks latent distances to match true-state distances up to a single learned global scale, $\mathcal{L}_{\text{metric}} = \mathbb{E}\big[(d_z - e^s\, d_x)^2\big]$ over sampled state pairs, where d_z, d_x are latent and true-state Euclidean distances. This supplies the missing isometry; we are explicit that it uses privileged true-state supervision (Section 6).

3.7 Protocol

The scientific unit is a three-seed sweep with mean $\pm$ standard error. Ablations and planning evals use 3 seeds $\times$ 12 episodes (planning) or 80-epoch trains (collapse ablation); real-data probes use 5-fold stratified cross-validation; the Tahoe probe uses 5 seeds $\times$ 300 epochs. Encoders are *frozen* before probing; planning is evaluated on held-out gLV seeds (sim_seed $= 10000 +$ seed). GPU runs were on a GB200 cluster; fast logic iteration on a CPU virtual environment.

4 Experiments and Results

4.1 Inverse dynamics: collapse and recovery

Hypothesis ([e], after [3]). A temporal JEPA collapses onto slow trajectory identity and discards the fast action; adding IDM should restore the decodability of the applied intervention while leaving slow-identity decodability (already saturated) unchanged.

Result ([m]). We freeze the encoder and fit ridge probes (with trajectory-level splits, no timepoint leakage) for the applied action, the one-step state change, the current state, and the initial state. In the *collapse-prone* regime (weak variance regularization: `std`=0.25, `cov`=1, `sim`=4) the IDM term sharply rescues action decodability (Table 1, Fig. 1): `fast_r2_action` rises from 0.520 ± 0.021 to 0.748 ± 0.051 (Δ+0.229), with on>off in *all three seeds* and non-overlapping error bars, while slow-identity decodability stays saturated ($\approx$ 0.99). In the *default* strong-VICReg regime the gap shrinks to +0.073 and is seed-noisy (one seed reverses). The IDM benefit therefore *substitutes* for variance regularization and is largest exactly where collapse is a real threat — a controlled, not over-claimed, conclusion.

Table 1: IDM collapse-and-recovery, induce-collapse regime (held-out R^2, mean±s.e., 3 seeds, 80 epochs). [M] (`ablation_collapse.json`).

probe (R^2)	IDM on	IDM off	Δ (on−off)
`fast_r2_action` (intervention)	$\mathbf{0.748 \pm 0.051}$	$\mathbf{0.520 \pm 0.021}$	+**0.229**
`fast_r2_delta` (one-step change)	0.819 ± 0.023	0.736 ± 0.012	+0.082
`fast_r2_state` (current state)	0.974 ± 0.003	0.963 ± 0.002	+0.011
`slow_r2_init` (identity, slow)	0.993 ± 0.001	0.989 ± 0.003	+0.005 (sat.)
`feat_std`	0.0107	0.0074	+0.0034

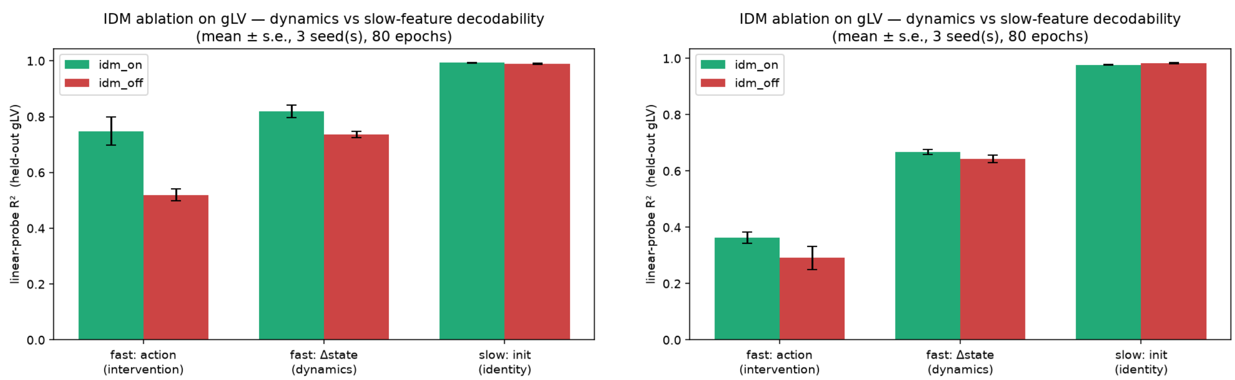

Figure 1: IDM ablation. **Left:** induce-collapse regime — IDM (green) strongly recovers action decodability over no-IDM (red); identity is saturated. **Right:** strong-VICReg default — the gap is small and overlaps. The IDM effect is regime-dependent.

4.2 Short-horizon forecasting: MDSINE2-style benchmark

On a hold-one-subject-out temporal benchmark (CLR-RMSE at horizons $h \in \{1, 3, 5, 10\}$, 3 seeds), the JEPA world model is the strongest forecaster at every horizon, beating persistence, a Ridge gLV-ℓ_2 baseline, and a strong gLV-net MLP (Fig. 2). This establishes that the latent predictor is genuinely *faithful* for short-horizon prediction — a fact that makes the planning failure in Section 4.4 all the more striking. (Code and measured JSON: `examples/microbiome_benchmark/`.)

4.3 Real-data downstream probe and the SIGReg/VICReg trade-off

We pretrain a static set-JEPA on a real microbial corpus and probe the frozen embedding on the Susagi infant-environment task with an apples-to-apples MLP probe and a corpus z-score. [M]: a SIGReg (LeJEPA) encoder at d=256 matches the published baseline (0.5255/0.8936 vs. 0.5270/0.8898 accuracy/AUC), and at d=384 beats it on both axes (0.5309/0.8986); a fine-tuned

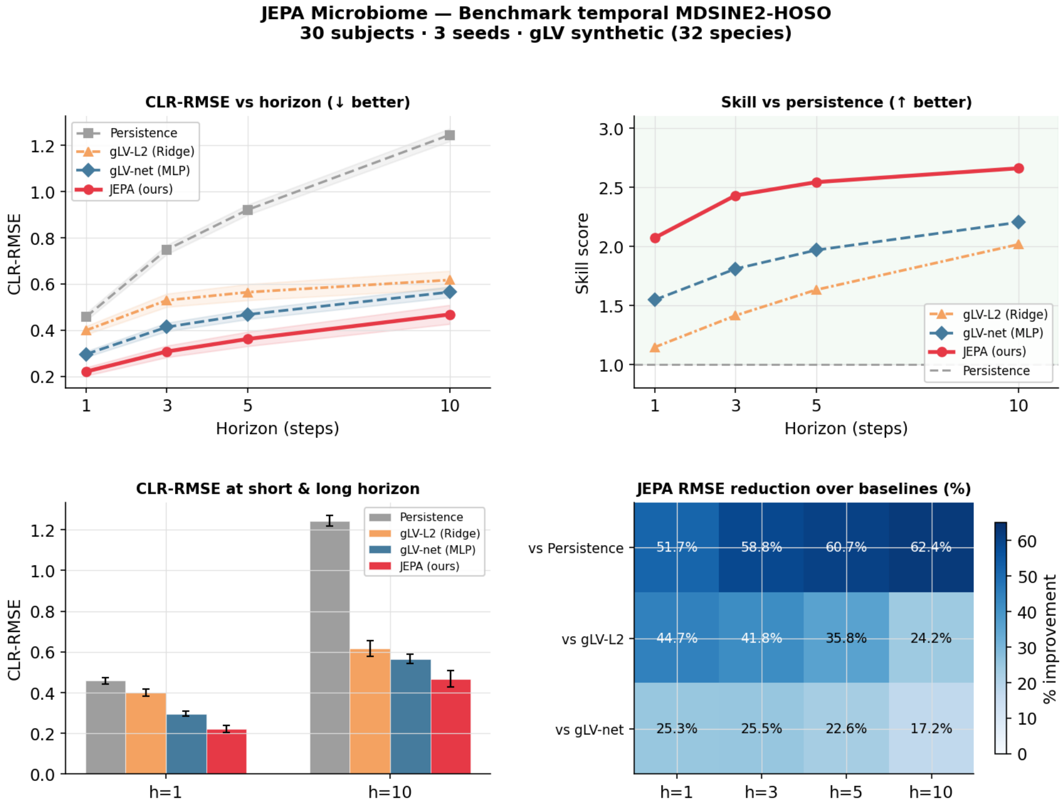

Figure 2: MDSINE2-style hold-one-subject-out benchmark: JEPA world model vs. persistence / gLV-ℓ_2 / gLV-net across horizons. [M]

upper bound (0.5899/0.9180) exceeds the published reference. A fair VICReg-$d128$ run sits just below baseline (0.4995/0.8884), and an EMA teacher consistently *hurts* (0.526→0.507 MLP).

Critically, the regularizer that *wins* the frozen-probe task is the *wrong* choice for planning. Replacing VICReg with SIGReg on the planning world model makes the embedding isotropic (`feat_std` 0.01→0.50) but drives the latent-vs-true distance correlation *negative* (Pearson −0.23) and inflates rollout error — isotropy is not the same as a plannable metric. This tension (isotropy helps probing, hurts planning) motivates the diagnosis below.

4.4 Planning: a diagnosed negative and its closure

This is the core of the paper. The task: steer the simulated community from one attractor to another, crossing a tolerance set at 15% of the inter-attractor distance.

(a) The task is solvable (oracle controllability, [m]). MPPI on the *true* gLV state reaches 100% success with final distance 0.790 when all $K{=}24$ species are actionable, and 0% for $K \leq 21$ (Fig. 3). Controllability is a property of the action budget, and the budget is adequate — so any downstream failure is the model's, not the task's.

(b) Pure-JEPA latent MPPI fails completely ([m]). Planning with the learned latent cost succeeds 0% of the time. The latent cost is *uninformative*: its rank correlation with true distance-to-target is Pearson −0.002, Spearman +0.02 — essentially noise — even though the *rollout* is faithful ($\approx 2\%$ one-step error). Faithful prediction, useless metric.

(c) Readout fidelity is not the bottleneck ([m]). Planning in a *decoded* state space improves with decoder quality: across two decoder settings (default-regularized, $R^2 \approx 0.78$, decoded-MPPI final 4.12; weakly-regularized, $R^2 \approx 0.89$, final 3.01), the stronger decoder's final distance beats a greedy baseline (3.26), yet planning *never crosses tolerance*. Better readout buys a better final distance, not success.

(d) Eight levers, still 0% ([m]). We attack the failure from every side and each is a clean negative: a learned monotonic cost head (rank correlation $0.71 \to 0.81$ with capacity, planning flat); an ensemble epistemic gate (uniform disagreement, not exploitable); on-policy (DAgger/MBPO-style) retraining (planner-trajectory error falls to ≈ 0.135, planning still 0%, final worsens); a multi-step free-running rollout objective (6-step error cut $0.611 \to 0.086$, -86%, planning still 0%). Dynamics fidelity is *exonerated*; the bottleneck is the geometry of the representation, not the predictor (Fig. 4).

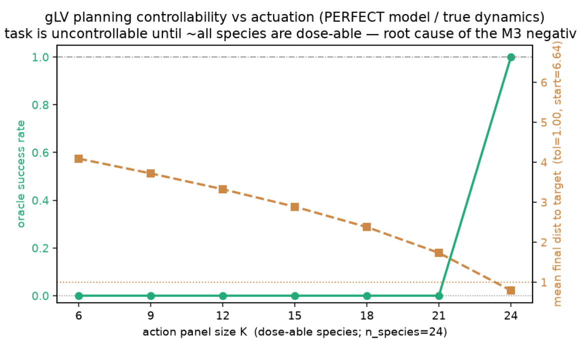

Figure 3: Oracle controllability vs. action budget K: 100% at K=24, 0% for $K \leq 21$. [M]

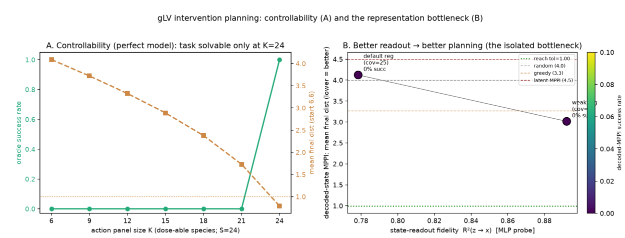

Figure 4: Layered diagnosis: controllable task + faithful rollout + *uninformative latent cost* $\Rightarrow$ representation geometry is the bottleneck. [M]

(e) Closing the loop: the metric auxiliary ([m]). Adding the isometry auxiliary of Section 3.6 (HYBRID, `metric_coeff`=0.3) turns raw-latent MPPI from 0% to **100%** $\pm$ **0**%, with final distance 0.804 — statistically indistinguishable from the oracle's 0.790 — while the latent-vs-true rank correlation jumps from 0.085 to **0.990** (Fig. 5). A sweep of `metric_coeff` shows a graceful trade-off: success erodes $100 \to 97 \to 92\%$ as the metric weight grows and the free-running rollout error rises, with 0.3 the sweet spot; community "recognition" (dominant-guild and basin identification) also improves. We are explicit that this is a *hybrid*, not a pure JEPA result: the auxiliary uses true gLV-state distances as supervision (Section 6).

(f) Generalization and falsification ([m]). The closure is not instance-specific: HYBRID at `metric_coeff`=0.3 crosses tolerance on 4/4 diverse gLV instances (pure JEPA 0% on all), matching the oracle (100%) on 2/4 and reaching 88.9%$\pm$7.3 on the other two. Two falsifiers confirm the metric must be *supplied*: a purely self-supervised IDM-reweighting drives the latent$\leftrightarrow$true correlation *negative* ($+0.085 \to -0.313$) and never plans; shrinking the bottleneck toward the true-state dimension nudges the correlation only marginally (≈ 0.2) and costs recognition — again 0%. A privileged metric is necessary and is not coercible from self-supervision alone on this testbed.

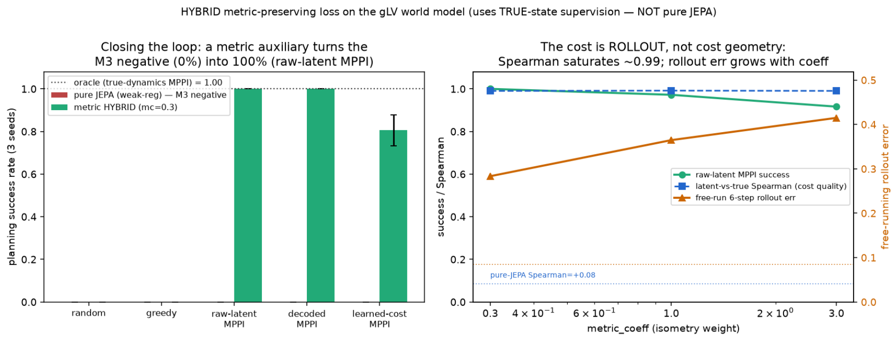

Figure 5: The isometry auxiliary closes the planning loop: pure JEPA 0% vs. HYBRID 100% at oracle quality, with latent↔true rank correlation 0.085→0.990. [M]

4.5 Sequencing-technology invariance

Should a microbiome embedding be invariant to the *sequencing technology* (amplicon vs. whole-genome) that produced it? We first measure a confounding ceiling: biology partially predicts technology (balanced accuracy ≈ 0.69), so perfect invariance would cost biology. Counter-intuitively, the *trained* JEPA is *more* technology-separable than its raw input (SIGReg 0.967 > VICReg 0.952 > raw 0.938 > random 0.897). We then test two removal strategies (Fig. 6). A domain-adversarial head (DANN) **fails**: technology accuracy never drops below 0.90 (never near the 0.69 floor) while biology is lost about three times faster. A deterministic moment-matching alignment (CORAL) gives a **partial** win: SIGReg+CORAL removes a linear technology leak (0.934→0.844, −0.090) at essentially flat biology (≈ 0.77), but an MLP probe still recovers technology at ≈ 0.96 — the fix is linear-only. Lesson: for a dominant, largely linear nuisance, *alignment beats adversarial*.

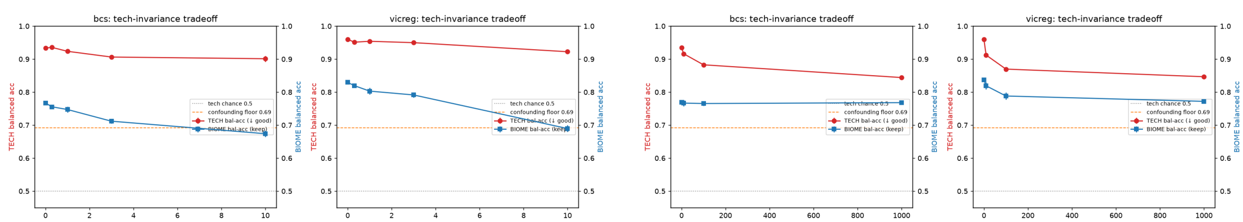

Figure 6: Technology-vs-biology trade-off. **Left:** DANN — biology is lost ∼ 3× faster than technology; no sweet spot. **Right:** CORAL — technology drops while biology stays essentially flat (a clearly favorable trade), but only the linear leak is removed. [M]

4.6 Real-world generality: Tahoe-100M drug perturbations

To test the recipe outside the simulator, we apply the action-conditioned predictor to real drug perturbations on frozen Tahoe-100M cell-state embeddings (the encoder is fixed, so there is no collapse to fight). [M]: the predictor *without* IDM beats a per-drug mean-shift baseline on held-out pairs (R^2_{shift} 0.201 vs. 0.149; the drug is decodable from the true shift at 75–84% top-1, chance ≈ 1%). Consistent with the collapse-specific reading of Section 4.1, IDM here *hurts* monotonically (0.201→0.017). Transfer to unseen cell lines fails (both arms collapse to around or below a no-op baseline). Single-step action-conditioning partially generalizes to real data; the IDM auxiliary does not, exactly as the collapse-specific hypothesis predicts (Fig. 7).

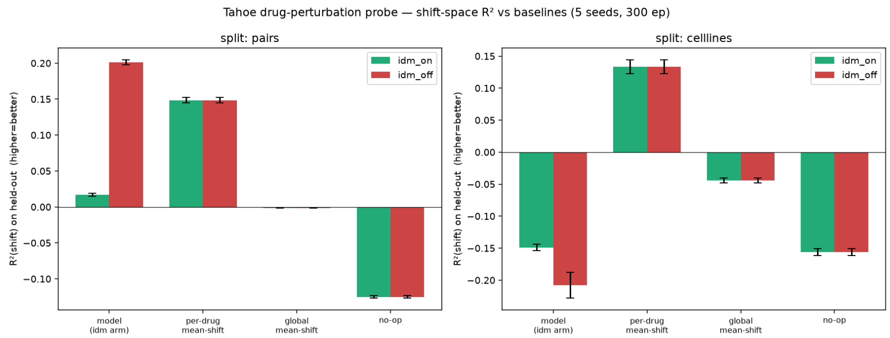

Figure 7: Tahoe drug-perturbation probe: the no-IDM predictor beats the per-drug baseline on held-out pairs (left); cross-cell-line transfer fails (right); IDM hurts on a frozen encoder. [M]

5 Discussion

Four lessons cut across the experiments. **(1) Faithful prediction is not a plannable metric.** A latent can give state-of-the-art short-horizon forecasts (Section 4.2) and a $\approx 2\%$ rollout error, yet carry a distance function that is pure noise for planning. The metric is a separate property and must be earned or supplied. **(2) The IDM benefit is collapse-specific.** It rescues fast features when variance regularization is weak (Section 4.1) and is redundant — even harmful — on a frozen encoder with no collapse (Section 4.6). **(3) Isotropy $\neq$ plannable geometry.** The SIGReg objective that wins the frozen probe breaks the planning metric (Section 4.3); "good for probing" and "good for planning" are different optima. **(4) For a dominant linear nuisance, alignment beats adversarial** (Section 4.5).

6 Limitations and integrity

The HYBRID is not a pure JEPA. The planning closure uses true gLV-state distances as supervision for the isometry term. It demonstrates that *a* metric closes the loop and that the bottleneck is geometric, but it is not a claim that pure self-supervision suffices — indeed our falsifiers show it does not (Section 4.4f). **The unseeded-decoder fluke.** An early decoded-planning run reported a 2.8% "success"; we traced it to an unseeded MLP decoder initialization, seeded it, and the result became a reproducible 0%. The corrected value is the one reported, and the fluke is recorded in the log rather than quietly dropped. **Simulator scope.** The planning results are on a synthetic gLV testbed designed to be controllable and non-monotonic; the metric closure need not transfer unchanged to real communities. **Real-data parsers** for the largest corpus were exercised on the cluster and are flagged unverified on commodity hardware. **Provenance gaps** (e.g. one missing baseline JSON, a Tahoe coefficient sweep whose intermediate log was not committed) are enumerated in `docs/WORK_REPORT.md`.

7 Conclusion

Energy-based JEPA world models can encode microbial dynamics well enough to forecast and, with the right auxiliary, to plan. The decisive insight is that a faithful predictive latent does not automatically furnish a metric in which planning works: across eight levers, pure-JEPA latent MPPI never crossed tolerance, and only an explicit isometry auxiliary — supplying the missing geometry

— closed the loop to oracle quality and generalized across instances. Inverse dynamics rescues fast, action-relevant features precisely when the representation is collapsing, and not otherwise. We hope the layered, honest methodology here — separating controllability, dynamics fidelity, readout, and metric — is useful for anyone trying to turn a predictive representation into a controller.

Acknowledgements / provenance. This work is built on the `eb_jepa` energy-based JEPA framework (`https://github.com/marinabar/eb_jepa`); the encoder, predictor, regularizers, and `unroll` machinery are upstream and credited under `LICENSE.md`. All experiments, results, and figures reproduced here are the author's contribution.

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